Introduction to Probability and Statistics

(EE 354 / CE 361)

Joint PMFs

1. $\mu_X = E[X] = \sum_x x \cdot p_X(x)$ *

2. Expected Value Rule

$Y = g(X)$

$$\mu_Y = E[Y] = E[g(X)] = \sum_x g(x) p_X(x)$$

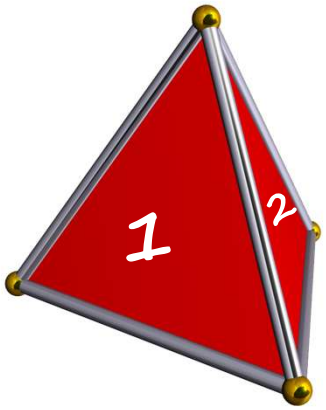
3. $\text{var}(X) = E[(X - E[X])^2]$

$f(x) = (X - E[X])^2$

$$\text{var}(X) = \sum_x (x - E[X]^*)^2 p_X(x)$$

↳ Constant $= \mu_X$

Exercise # 1

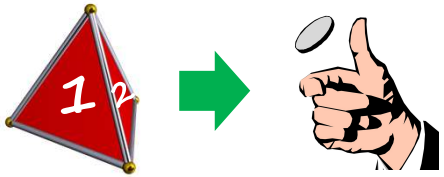


Equally likely

- You throw a tetrahedral dice 12 times.

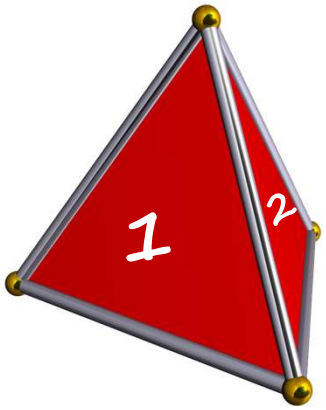
R.V X is the number of times you get 3 in the 12 throws & $P_X(k)$ is the Probability of getting No 3, k-times in 12 throws

What is $P_X(5)$ =



parameters $p = 1/4$

Exercise # 1

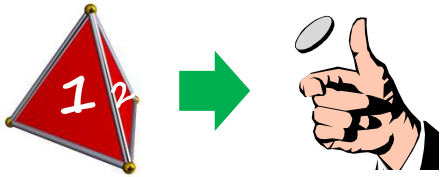


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What is $P_X(5)$ =



parameters p = 1/4

N = 12, p = ¼ , 1-p = ¾ , k=5

$$p_X(5) = \binom{12}{5} \cdot \left(\frac{1}{4}\right)^5 \cdot \left(\frac{3}{4}\right)^7$$

Exercise No 02!

A graduating CS student of HU goes to career fair booths in technology sector - & his/her likelihood of receiving an off-campus interview invitation after a career fair booth visit depends on how well he/she did in EE 354 course. Specifically, an A in the course results in a probability $p_1 = 0.95$ of obtaining an invitation, whereas a C in EE 354 course results in a probability of $p_2 = 0.15$ of an invitation. Assuming that each student visits 5 booths during a typical career fair, find the probability that an A grade student in EE 354 course will not get an off-campus interview invitation. Similarly, find the probability that a C grade student in EE 354 course will get an invitation during a typical career fair

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P(an A grade student does not get an invitation in 5 trials)= $\sum_{k=6}^{\infty}(1-p_1)^{k-1.}p_1 = (1-p_1)^5$

P(C grade student gets an invitation in 5 trials)= $1-(1-p_2)^5$

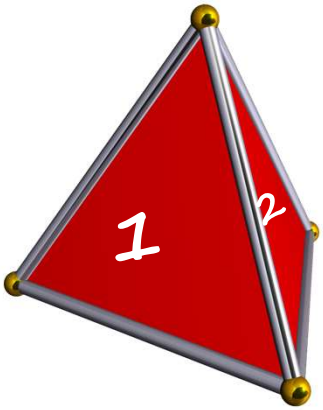
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P(an A grade student does not get an invitation in 5 trials)= $\sum_{k=6}^{\infty}(1-p_1)^{k-1.}\,p_1 = (1-p_1)^5$
$= (1-0.95)^5 = (0.05)^5$

P(C grade student gets an invitation in 5 trials)= $1-(1-p_2)^5 = (1-0.15)^5 = (0.85)^5$

Exercise # 3



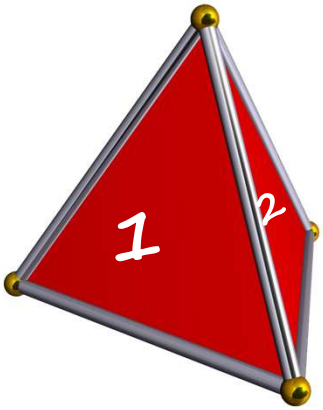
Equally likely

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \ldots; X_n)$, where $X_{i's}$ are outcomes of identical and independently distributed tetrahedral die (4-sided). As an example $Z_2 = \min(X_1;X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

1. What is the $P(Z_{10} > 1)$?

2. What is the $P(Z_{10} = 3)$?

Exercise # 3



Equally likely

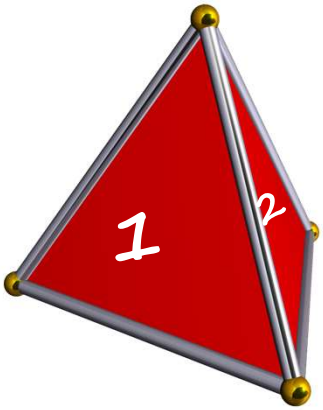
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1. What is the $P(Z_{10} > 1)$?

$$P(Z_{10} > 1) = \left(\frac{3}{4}\right)^{10}$$

2. What is the $P(Z_{10} = 3)$?

Exercise # 3



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1. What is the $P(Z_{10} > 1)$?

$$P(Z_{10} > 1) = \left(\frac{3}{4}\right)^{10}$$

We define two events
A - Either 3 or 4 appears in ALL ten throws,
B - 4 appears in ALL ten throws

2. What is the $P(Z_{10} = 3)$?

$$P(A) = \left(\frac{2}{4}\right)^{10} \qquad P(B) = \left(\frac{1}{4}\right)^{10}$$

$$P(Z_{10} = 3) = \left(\frac{2}{4}\right)^{10} - \left(\frac{1}{4}\right)^{10}$$

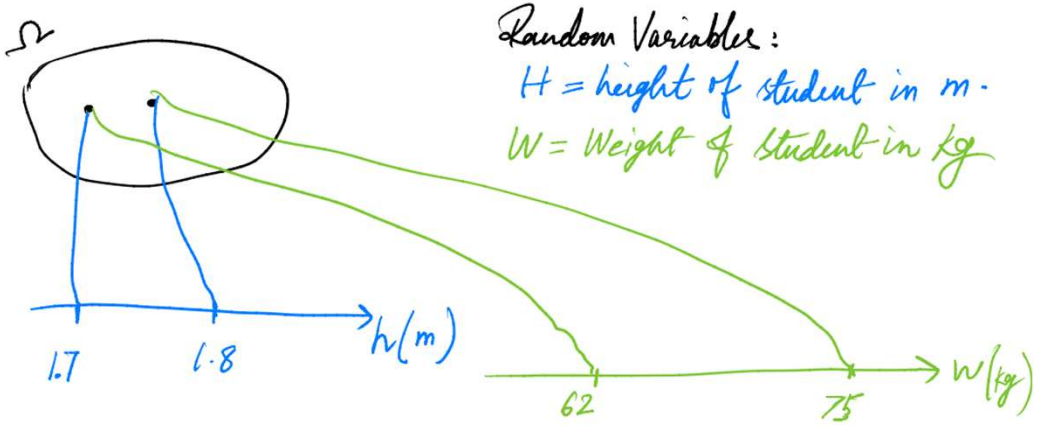
Today's Agenda

- Joint PMFs
  - Definition/Intuition
  - Properties
  - Extension to more than Two Random Variables
  - Mean
  - Mean of Binomial Distribution

Multiple Random Variables on the same Sample Space

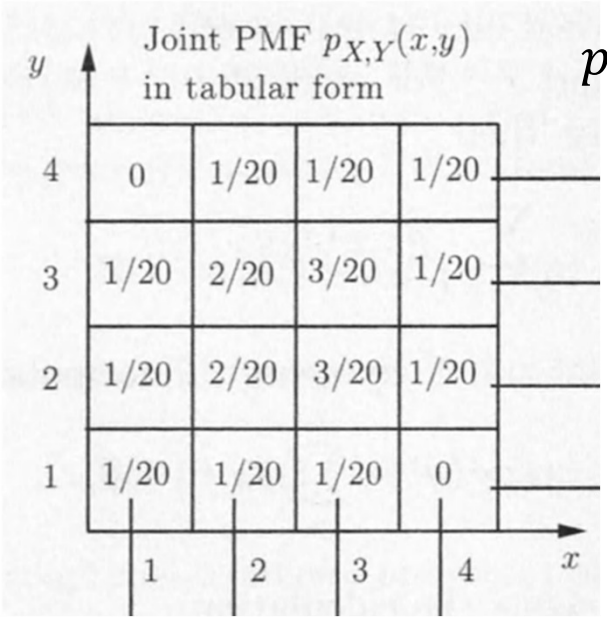
- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class

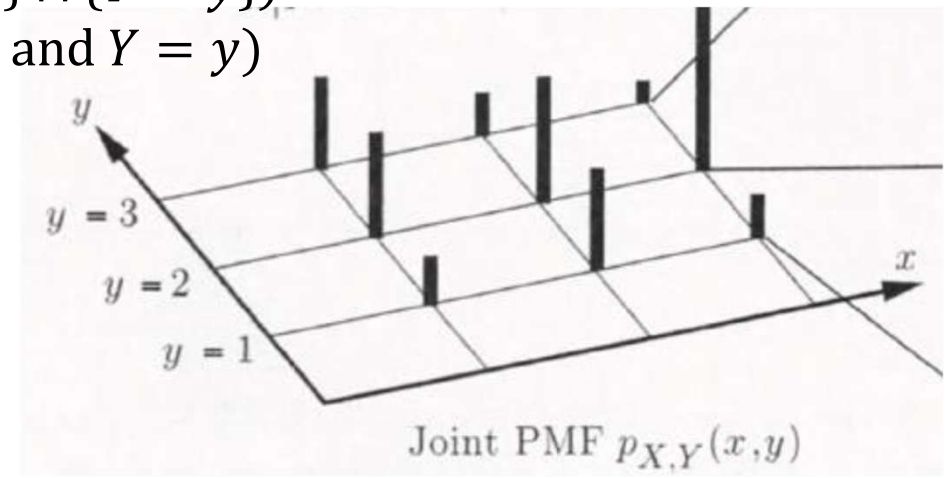


Multiple Random Variables and Joint PMF

- We can extend the concepts of PMF and expectation developed so far for a single RV to multiple RVs
- Consider two discrete random variables X and Y associated with the same experiment.
  - The probabilities of the values that X and Y can take are captured by the joint PMF of X and Y, denoted by $p_{X,Y}$
  - In particular, if (x, y) is a pair of possible values of X and Y, the probability mass of (x, y) is the probability of the event $\{X = x, Y = y\}$



$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y)$$
$$= \mathbf{P}(\{X = x\} \cap \{Y = y\})$$
$$= \mathbf{P}(X = x \text{ and } Y = y)$$



Properties of Joint PMF

$$\boxed{\sum_x \sum_y p_{X,Y}(x,y) = 1}$$

- We can calculate the PMFs of X and Y from Joint PMFs:

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

The formula for $p_X(x)$ can be verified using the calculation

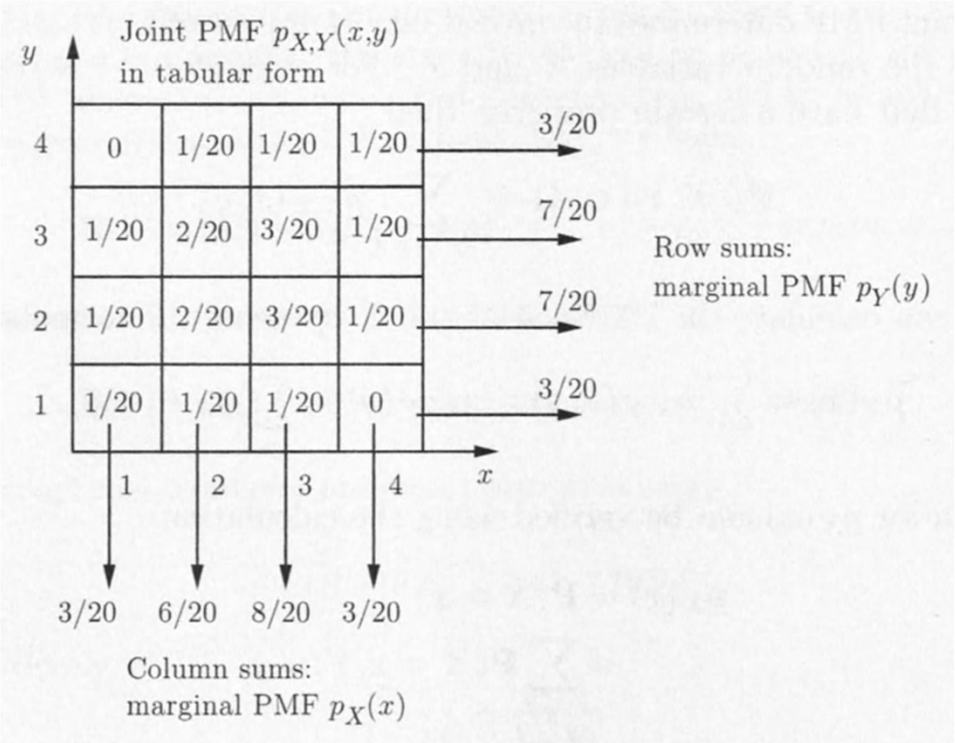
$$\begin{aligned} p_X(x) &= \mathbf{P}(X = x) \\ &= \sum_y \mathbf{P}(X = x, Y = y) \\ &= \sum_y p_{X,Y}(x,y), \end{aligned}$$

where the second equality follows by noting that the event $\{X = x\}$ is the union of the disjoint events $\{X = x, Y = y\}$ as y ranges over all the different values of Y. The formula for $p_Y(y)$ is verified similarly. We sometimes refer to p_X and p_Y as the **marginal** PMFs, to distinguish them from the joint PMF.

Properties of Joint PMF

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$



Joint PMF: More than Two Random Variables

- The joint PMF of three random variables X, Y, and Z is defined as:

$$p_{X,Y,Z}(x, y, z) = \mathbf{P}(X = x, Y = y, Z = z).$$

- Also,

$$\sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$$

- Corresponding marginal PMFs can also be analogously obtained:

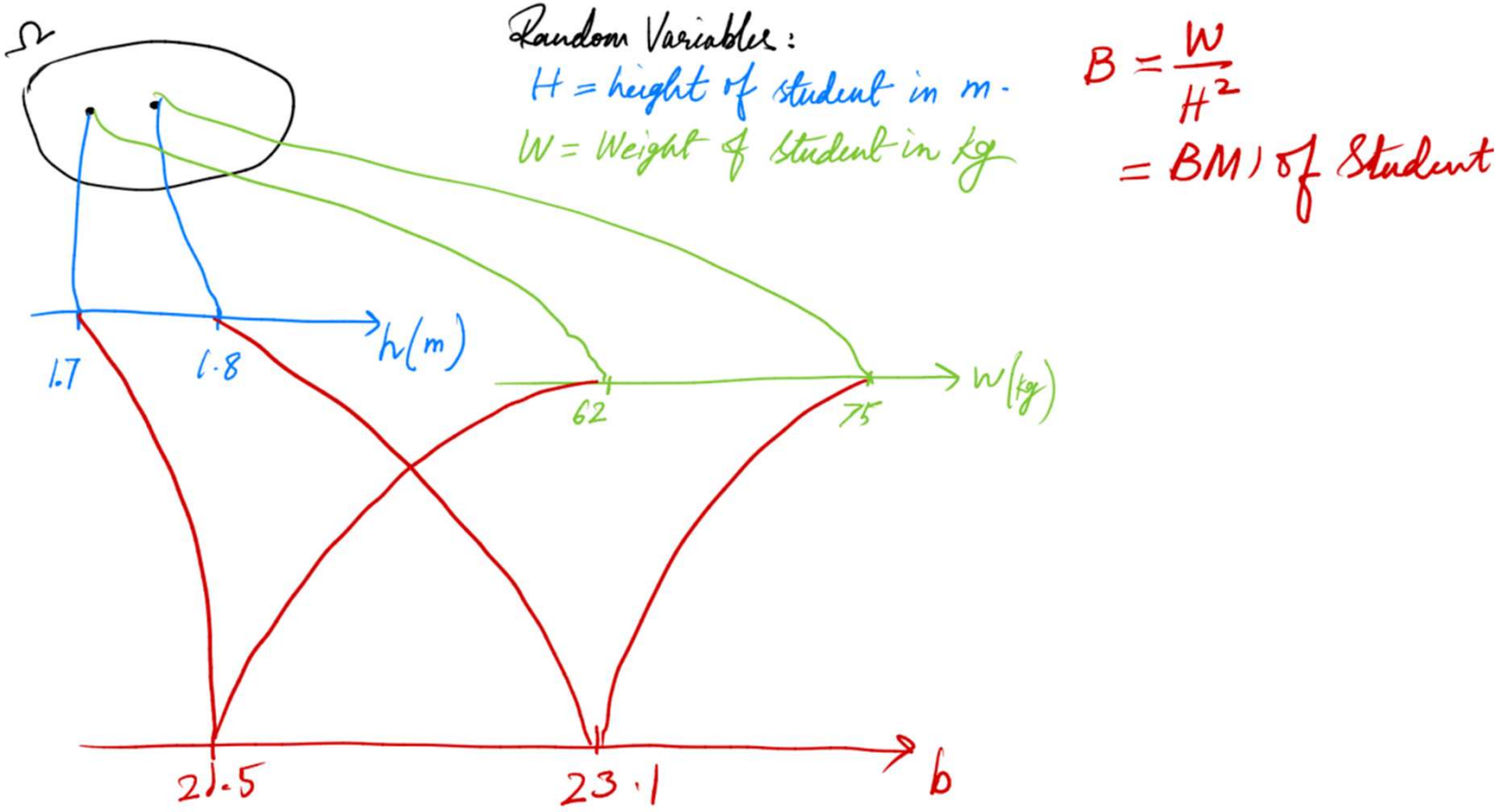
$$p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z),$$

and

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x, y, z).$$

Remember: Functions of Random Variables

- A function of one or several random variables is also a random variable.
- Meaning of $X + Y$: Our RV takes on value $(x + y)$ when X takes on value x and Y takes on value y



Functions of Random Variables: Context/Utilization

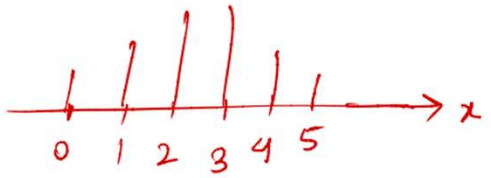
X : No. of red signals out of 5 on my commute

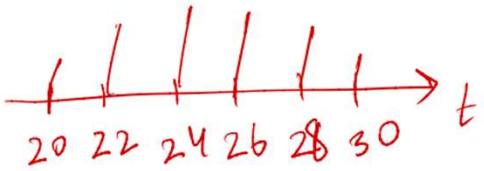
T : Commute time

$$T = 20 + 2X$$

$$E[X] = \sum_x x\, p_X(x)$$

$$\mu_X \,,\ \mu_T = 20 + 2\mu_X$$





Functions of Multiple Random Variables

- We can extend the concepts of PMF and expectation developed so far for functions of a single RV to functions of multiple RVs.

- If $Z = g(X, Y)$ is a function of random variable X and Y, then Z is also a random variable and its PMF can be calculated from the joint PMF $p_{X,Y}$ as follows:

$$p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{\{(x,y) \mid g(x,y)=z\}} p_{X,Y}(x, y).$$

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- Expected Value Rule:

$$\mathbf{E}\big[g(X, Y)\big] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y).$$

- In the special case where g is linear and of the form $aX + bY + c$, where a, b, and c are given scalers, we have:

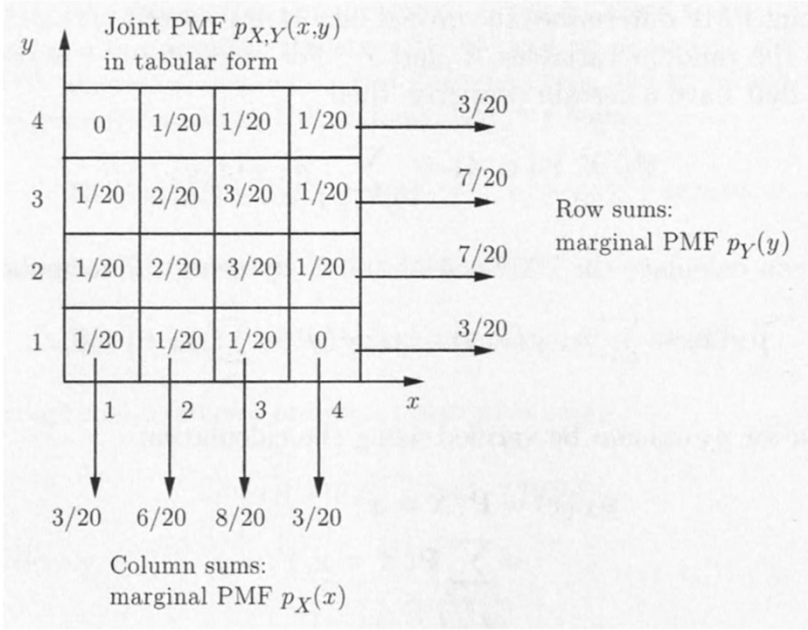
$$\mathbf{E}[aX + bY + c] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c.$$

Example: Functions of Multiple Random Variables

- Consider our previous example:
- Consider a new random variable Z:

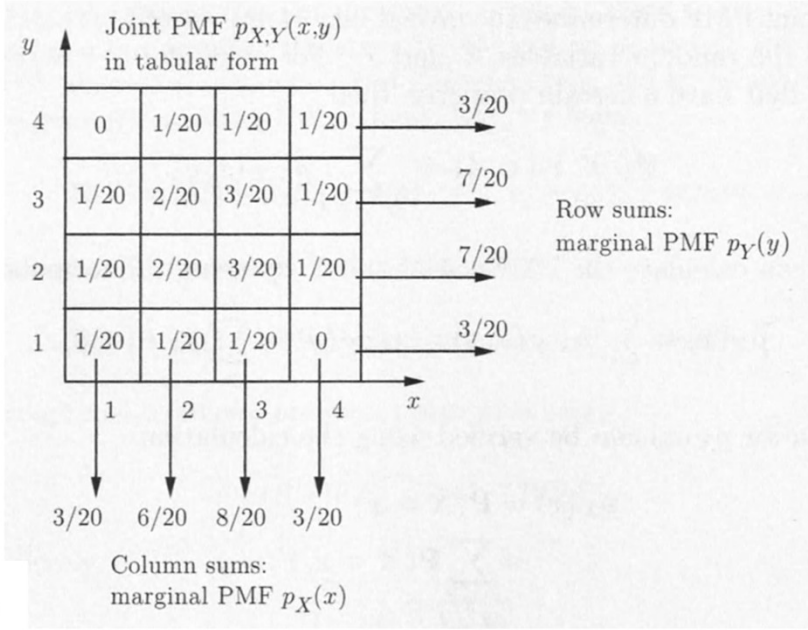
$$Z = X + 2Y$$

- What is the PMF of Z?



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$$p_Z(z) = \sum_{\{(x,y)\,|\,x+2y=z\}} p_{X,Y}(x,y),$$

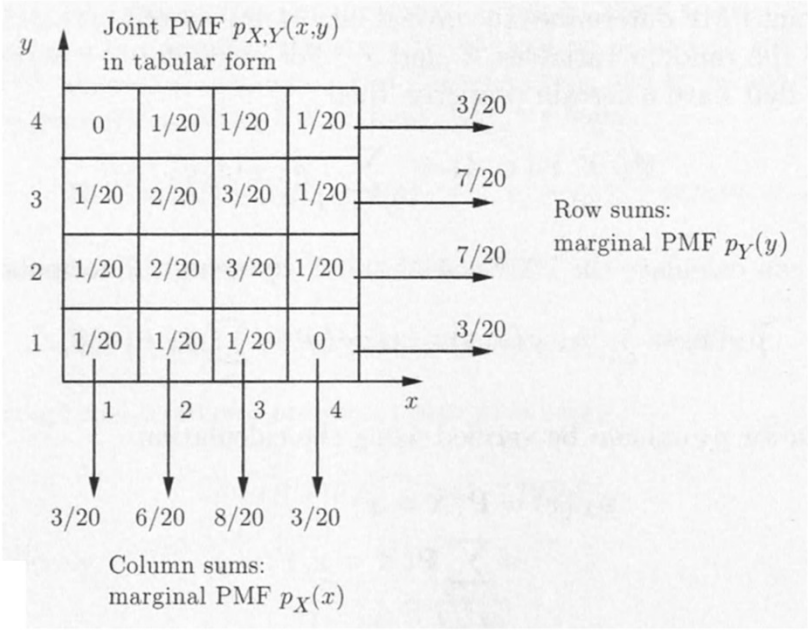
$$p_Z(3) = \frac{1}{20},\quad p_Z(4) = \frac{1}{20},\quad p_Z(5) = \frac{2}{20},\quad p_Z(6) = \frac{2}{20},\quad p_Z(7) = \frac{4}{20},$$

$$p_Z(8) = \frac{3}{20},\quad p_Z(9) = \frac{3}{20},\quad p_Z(10) = \frac{2}{20}.\quad p_Z(11) = \frac{1}{20},\quad p_Z(12) = \frac{1}{20}.$$

$$\begin{aligned}\mathbf{E}[Z] &= \sum z p_Z(z) \\ &= 3\cdot\frac{1}{20} + 4\cdot\frac{1}{20} + 5\cdot\frac{2}{20} + 6\cdot\frac{2}{20} + 7\cdot\frac{4}{20} \\ &\quad + 8\cdot\frac{3}{20} + 9\cdot\frac{3}{20} + 10\cdot\frac{2}{20} + 11\cdot\frac{1}{20} + 12\cdot\frac{1}{20} \\ &= 7.55.\end{aligned}$$

Example: Functions of Multiple Random Variables

- Consider our previous example:



- Consider a new random variable Z:

$$Z = X + 2Y$$

- What is the PMF of Z?

$$p_Z(z) = \sum_{\{(x,y)\,|\,x+2y=z\}} p_{X,Y}(x,y),$$

$$p_Z(3) = \frac{1}{20},\quad p_Z(4) = \frac{1}{20},\quad p_Z(5) = \frac{2}{20},\quad p_Z(6) = \frac{2}{20},\quad p_Z(7) = \frac{4}{20},$$

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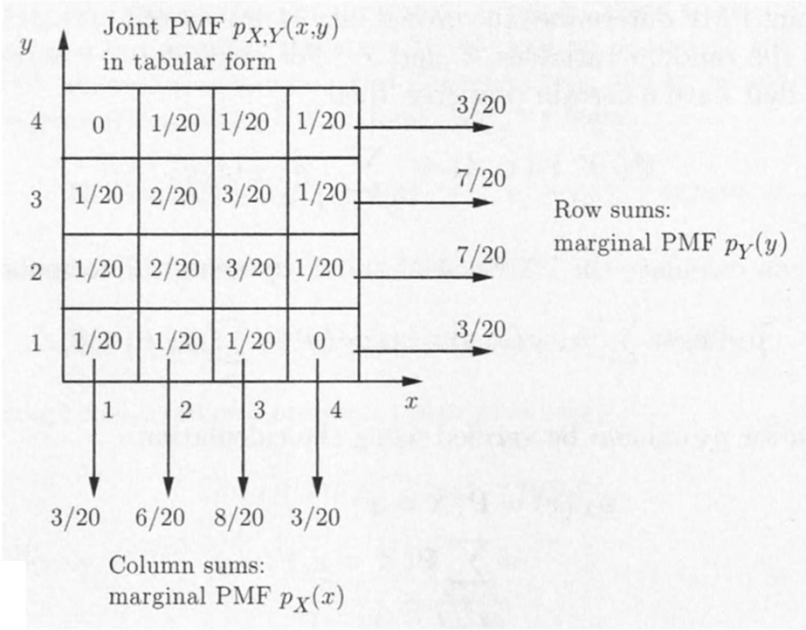
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- We can refer to the fact that Z is a **linear function** of X and Y

$$\mathbf{E}[Z] = \mathbf{E}[X] + 2\mathbf{E}[Y].$$

$$\mathbf{E}[X] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{6}{20} + 3 \cdot \frac{8}{20} + 4 \cdot \frac{3}{20} = \frac{51}{20},$$

$$\mathbf{E}[Y] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{7}{20} + 3 \cdot \frac{7}{20} + 4 \cdot \frac{3}{20} = \frac{50}{20},$$

$$\mathbf{E}[Z] = \frac{51}{20} + 2 \cdot \frac{50}{20} = 7.55.$$

Extensions to More than Two Random Variables

if g is linear and has the form $aX + bY + cZ + d$, then

$$\mathbf{E}[aX + bY + cZ + d] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c\mathbf{E}[Z] + d.$$

Furthermore, there are obvious generalizations of the above to more than three random variables. For example, for any random variables $X_1, X_2, \ldots, X_n$ and any scalars $a_1, a_2 \ldots, a_n$, we have

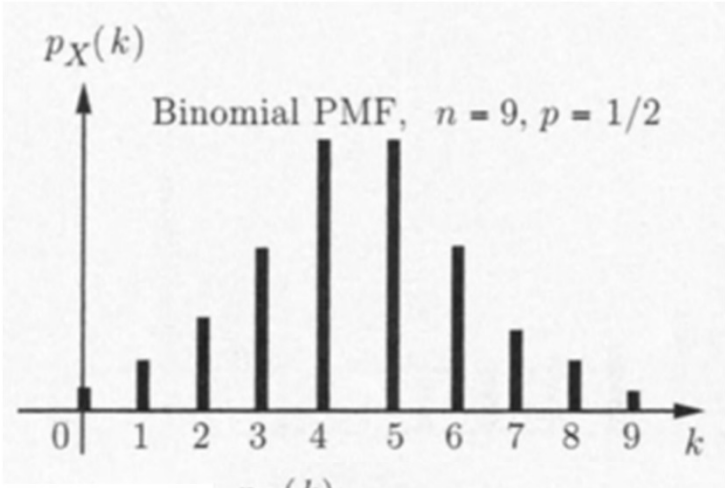
$$\mathbf{E}[a_1X_1 + a_2X_2 + \cdots + a_nX_n] = a_1\mathbf{E}[X_1] + a_2\mathbf{E}[X_2] + \cdots + a_n\mathbf{E}[X_n].$$

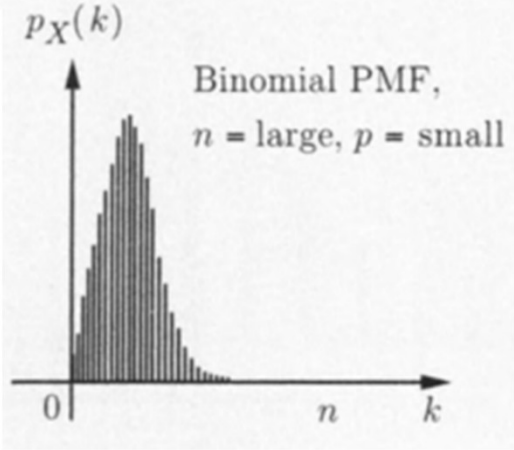
- Example

  $\mathbf{E}[2X + 3Y - Z] = \mathbf{E}[2X] + \mathbf{E}[3Y] - \mathbf{E}[Z] = 2\mathbf{E}[X] + 3\mathbf{E}[Y] - \mathbf{E}[Z]$

Binomial Random Variable

- Experiment
  - n independent tosses of a coin where at each toss, the coin comes up a head with probability p and a tail with probability $(1 - p)$.
- Sample Space
  - Set of Head (H) and Tail (T) sequences of length n
- Random Variable
  - The number of Heads in the n-toss sequence
- PMF
  - $\boldsymbol{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \ldots, n}$
- Model of:
  - Number of successes in a given number of independent trials
- Parameters: $\boldsymbol{n \in \mathbb{Z}^{++}}$, $\boldsymbol{p \in [0, 1]}$





Mean of a Binomial Random Variable

- Random Variable
  - The number of Heads in the n-toss sequence
- PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \dots, n$$

- Mean

$$\mathbf{E}[X] = \sum_{k=0}^{n} k p_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

  - An alternative approach:

Lets consider n random variables $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

$X_1, X_2, \dots, X_n$ are Bernoulli random variables with common mean p
Their sum

$$X = X_1 + X_2 + \cdots + X_n$$

is a binomial random variable with parameters n and p.

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n]$$

$$\mathbf{E}[X] = np$$

Example: Mean of a Binomial Random Variable

- Example
  - Your probability class has 60 students and each student has probability $\frac{1}{3}$ of getting an A, independent of any other student. Consider the RV:

    X = The number of students that get an A
  - What is the mean of X?

Example: Mean of a Binomial Random Variable

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  - Your probability class has 60 students and each student has probability $\frac{1}{3}$ of getting an A, independent of any other student. Consider the RV:

    X = The number of students that get an A
  - What is the mean of X?

Note that X is a binomial RV.

$$\mathbf{E}[X] = np$$

$$\mathbf{E}[X] = 60\left(\frac{1}{3}\right) = 20$$